

TCS3351 CRYPTOGRAPHY AND DATA SECURITY
TUTORIAL 2
MATHEMATICAL BACKGROUND – I

1. a) Which of the following relations are true and which are false?
(i) $5 \mid 26$ (ii) $3 \mid 123$ (iii) $27 \nmid 127$ (iv) $15 \nmid 21$
b) Find the results of the following operations.
(i) $-78 \bmod 13$ (ii) $0 \bmod 13$ (iii) $150 \bmod 15$ (iv) $-66 \bmod 33$
2. a) Using the Euclidean algorithm, find the greatest common divisor of the pairs on integers, 88 and 220.
b) Using the extended Euclidean algorithm, find the greatest common divisor of the pairs (4,7) and the value of s and t.
3. Perform the following operations using reduction first.
(i) $(4223 + 17323) \bmod 10$
(ii) $(2468 - 43) \bmod 15$
(iii) $(144 \times 34) \bmod 12$
(iv) $(221 \times 23) \bmod 22$
4. Check whether the multiplicative inverse exists for each of the following integers and if exists, find it using the extended Euclidean algorithm.
(i) 61 in $\mathbf{Z}_{90}$ (ii) 132 in $\mathbf{Z}_{180}$
5. Let us assign numeric values to the uppercase alphabet (A=0, B=1, ..., Z=25). Answer the following using modulo 26.
(i) $(A+N) \bmod 26$ (ii) $(A+6) \bmod 26$
(ii) $(Y-5) \bmod 26$ (iv) $(C-10) \bmod 26$

References:

- 1) Behrouz A. Forouzan, Debdeep Mukhopadhyay, "Cryptography and Network Security", Third Edition, McGraw-Hill, 2015.
- 2) William Stallings, "Cryptography and Network Security – Principles and Practice", Seventh Edition, Prentice Hall, 2017.

Extra Exercises:

1.9. Use the Euclidean algorithm to compute the following greatest common divisors.

- (a) $\gcd(291, 252)$.
- (b) $\gcd(16261, 85652)$.
- (c) $\gcd(139024789, 93278890)$.
- (d) $\gcd(16534528044, 8332745927)$.

1.17. Do the following modular computations. In each case, fill in the box with an integer between 0 and $m - 1$, where m is the modulus.

- (a) $347 + 513 \equiv \boxed{} \pmod{763}$.
- (b) $3274 + 1238 + 7231 + 6437 \equiv \boxed{} \pmod{9254}$.
- (c) $153 \cdot 287 \equiv \boxed{} \pmod{353}$.
- (d) $357 \cdot 862 \cdot 193 \equiv \boxed{} \pmod{943}$.
- (e) $5327 \cdot 6135 \cdot 7139 \cdot 2187 \cdot 5219 \cdot 1873 \equiv \boxed{} \pmod{8157}$.
(*Hint.* After each multiplication, reduce modulo 8157 before doing the next multiplication.)
- (f) $137^2 \equiv \boxed{} \pmod{327}$.
- (g) $373^6 \equiv \boxed{} \pmod{581}$.
- (h) $23^3 \cdot 19^5 \cdot 11^4 \equiv \boxed{} \pmod{97}$.

1.18. Find all values of x between 0 and $m - 1$ that are solutions of the following congruences. (*Hint.* If you can't figure out a clever way to find the solution(s), you can just substitute each value $x = 1, x = 2, \dots, x = m - 1$ and see which ones work.)

- (a) $x + 17 \equiv 23 \pmod{37}$.
- (b) $x + 42 \equiv 19 \pmod{51}$.

1.32. For each of the following primes p and numbers a , compute $a^{-1} \pmod{p}$

- (a) $p = 47$ and $a = 11$.
- (b) $p = 587$ and $a = 345$.
- (c) $p = 104801$ and $a = 78467$.

References:

- 1) Jeffrey Hoffstein, Jill Pipher and Joseph H. Silverman, "An Introduction to Mathematical Cryptography", Second Edition, Springer, 2014.